Parishram

2026 Class 12th Commerce

Lecture No.-2

Mathematics(Core)

Integrals

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Recap *of previous lecture*

1. Integral

Topics *to be covered*

1. Integrals

Ex. $\underline{\underline{7.1}}$

11. $\int \frac{x^3+5x^2-4}{x^2}\,dx$

$$\int \frac{x^3}{x^2}+\frac{5x^2}{x^2}-\frac{4}{x^2}\,dx$$

$$= \int \left(x^1 + 5 - 4x^{-2}\right) dx$$

$$= \frac{x^2}{2} + 5x - \frac{4x^{-1}}{-1} + k$$

$$= \frac{x^2}{2} + 5x + \frac{4}{x} + k$$

12. $\int \frac{x^3+3x+4}{\sqrt{x}}\,dx$

$$\int \frac{x^3}{(x)^{1/2}} + \frac{3x^1}{(x)^{1/2}} + \frac{4}{(x)^{1/2}}\,dx$$

$$= \int (x)^{\frac{5}{2}} + 3(x)^{1/2} + 4(x)^{-1/2}\,dx$$

$$= \frac{(x)^{7/2}}{7/2} + \frac{3(x)^{3/2}}{3/2} + \frac{4(x)^{1/2}}{1/2} + K$$

13. $\int \frac{x^3 - x^2 + x - 1}{x-1} dx$

$$\int \frac{\overline{x^3 - x^2} + \overline{x - 1}}{x-1} dx$$

$$= \int \frac{x^2 \overline{(x-1)} + 1 \overline{(x-1)}}{(x-1)} dx$$

$$= \int \frac{(x-1)[x^2+1]}{(x-1)} dx$$

$$= \int x^2 + 1 \, dx$$

$$= \frac{x^3}{3} + x + k$$

15. $\int \sqrt{x}(3x^2 + 2x + 3)\, dx$

$$\int (x)^{\frac{1}{2}}\left[3x^2+2x+3\right]dx$$

$$= \int\left[3(x)^{\frac{1}{2}}\cdot x^2 + 2(x)^{\frac{1}{2}}\cdot x^1 + 3(x)^{\frac{1}{2}}\right]dx$$

$$= \int\left[3(x)^{\frac{5}{2}} + 2(x)^{3/2} + 3(x)^{\frac{1}{2}}\right]dx$$

$$= \frac{3(x)^{7/2}}{7/2} + \frac{2(x)^{5/2}}{5/2} + \frac{3(x)^{3/2}}{3/2} + k$$

16. $\int (2x - 3\cos x + e^x)\, dx$

$$\int\left[2x - 3\cos x + e^x\right]dx$$

$$= 2\left(\frac{x^2}{2}\right) - 3(\sin x) + e^x + k$$

17. $\int (2x^2 - 3\sin x + 5\sqrt{x})\, dx$

18. $\int \sec x\,(\sec x + \tan x)\, dx$

$\int \left[2\underbrace{x^2} - 3\underbrace{\sin x} + 5(x)^{\frac{1}{2}}\right] dx$

$= \frac{2x^3}{3} - 3(-\cos x) + \frac{5(x)^{3/2}}{3/2} + k$

$\int \sec x\,(\sec x + \tan x)\, dx$

$= \int \left[\sec^2 x + \underbrace{\sec x \tan x}\right] dx$

$= \tan x + \sec x + k$

19. $\int \frac{\sec^2 x}{\operatorname{cosec}^2 x} dx$

20. $\int \frac{2 - 3\sin x}{\cos^2 x} dx.$

$$= \int \frac{\sec^2 x}{\operatorname{cosec}^2 x} dx$$

$$= \int \frac{\sin^2 x}{\cos^2 x} dx$$

$$= \int \underbrace{\tan^2 x} dx$$

$$= \int (\sec^2 x - 1) dx$$

$$= \tan x - x + K$$

(20)

$$= \int \frac{2}{\cos^2 x} - \frac{3\sin x}{\cos^2 x} dx$$

$$= \int \left[2\sec^2 x - 3\underbrace{\tan x \cdot \sec x}\right] dx$$

$$= 2\tan x - 3\sec x + K$$

.22. If $\frac{d}{dx}f(x) = 4x^3 - \frac{3}{x^4}$ such that $f(2) = 0$. Then $f(x)$ is

(A) $x^4 + \frac{1}{x^3} - \frac{129}{8}$ ✓

~~(B)~~ $x^3 + \frac{1}{x^4} + \frac{129}{8}$

(C) $x^4 + \frac{1}{x^3} + \frac{129}{8}$

~~(D)~~ $x^3 + \frac{1}{x^4} - \frac{129}{8}$

Sol:

$$f(x) \underset{I}{\overset{D}{\rightleftarrows}} 4x^3 - \frac{3}{x^4}$$

$$f(x) = \int\left(4x^3 - \frac{3}{x^4}\right)dx$$

$$= \int\left(4x^3 - 3x^{-4}\right)dx$$

$$= \frac{4x^4}{4} - \frac{3x^{-3}}{-3} + k$$

$$f(x) = x^4 + \frac{1}{x^3} + k$$

now $f(2) = 0$

$$(2)^4 + \frac{1}{(2)^3} + k = 0$$

$$16 + \frac{1}{8} + k = 0$$

$$\frac{129}{8} + k = 0$$

$$\boxed{k = -\frac{129}{8}}$$

$$f(x) = x^4 + \frac{1}{x^3} - \frac{129}{8}$$

$\int x^n dx = \frac{x^{n+1}}{n+1} + k$

$\int \frac{1}{x} dx = \log|x| + k$

$\int e^x dx = e^x + k$

$\int (a)^x dx = \frac{a^x}{\log a} + k$

$$a^x \xrightarrow{D} a^x \cdot \log a$$

$$\frac{a^x}{\log a} \xrightarrow{D} \frac{a^x \cdot \cancel{\log a}}{\cancel{\log a}}$$

Eg $\int (2)^x dx$

$= \frac{(2)^x}{\log 2} + k$

Eg $\int \frac{5^x}{7^x} dx$

$= \int \left(\frac{5}{7}\right)^x dx$

$= \frac{\left(\frac{5}{7}\right)^x}{\log\left(\frac{5}{7}\right)} + k$

Q $\int \frac{(a^x + b^x)^2}{a^x b^x} dx$

$$= \int \frac{a^{2x}}{a^x b^x} + \frac{b^{2x}}{a^x b^x} + 2\frac{a^x b^x}{a^x b^x} dx$$

$$= \int \left[\frac{a^x}{b^x} + \frac{b^x}{a^x} + 2\right] dx$$

$$= \int \left(\frac{a}{b}\right)^x + \left(\frac{b}{a}\right)^x + 2 \, dx$$

$$= \frac{\left(\frac{a}{b}\right)^x}{\log\left(\frac{a}{b}\right)} + \frac{\left(\frac{b}{a}\right)^x}{\log\left(b/a\right)} + 2x + K$$

QUESTION

$\int e^x dx = e^x + k$

$\int e^{\underbrace{ax+b}_{\text{Linear}}} dx = \dfrac{e^{ax+b}}{a} + k$

Q $\int (e)^{2x} dx$

$= \dfrac{(e)^{2x}}{2} + k$

Q $\int (e)^{4x+3} dx$

$= \dfrac{(e)^{4x+3}}{4} + k$

Q $\int (e)^{3-5x} dx$

$= \dfrac{e^{3-5x}}{-5} + k$

Q $\int \dfrac{1}{e^{4x}} dx$

$= \int e^{-4x} dx$

$= \dfrac{e^{-4x}}{-4} + k$

$\int \frac{1}{x}\,dx = \log|x| + k$

$$\int \frac{1}{ax+b} = \frac{\log|ax+b|}{a} + k$$

Linear

eg $\int \frac{1}{3x+4}\,dx$

$= \frac{\log|3x+4|}{3} + k$

eg $\int \frac{1}{5-3x}\,dx$

$= \frac{\log|5-3x|}{-3} + k$

eg $\int \frac{1}{1+x^2}\,dx$

$= \cancel{\frac{\log|1+x^2|}{2x}} + k$

$= \tan^{-1}x + k$

QUESTION

$$\int \cos(x)\,dx = \sin x + k$$

$$\int \cos\underbrace{(2x)}_{\text{Linear}}\,dx = \frac{\sin(2x)}{2} + k$$

$$\int \sin(4x+5)\,dx = -\frac{\cos(4x+5)}{4} + k$$

$$\int \sec^2(3x+7)\,dx = \frac{\tan(3x+7)}{3} + k$$

QUESTION

$$\int (x)^5 \, dx$$

$$= \frac{x^6}{6} + k$$

$$\int x^{10} \, dx$$

$$= \frac{x^{11}}{11} + k$$

$$\int (\underbrace{2x+3}_{\text{Linear}})^{20} \, dx$$

$$= \frac{(2x+3)^{21}}{(21)(2)} + k$$

Q $\int (4-5x)^6 \, dx$

$$= \frac{(4-5x)^7}{(7)(-5)} + k$$

6. $\int (4\, e^{3x} + 1)\, dx$

$= \int 4 \underbrace{(e)}^{3x} + 1 \quad dx$

$= \dfrac{4\,(e)^{3x}}{3} + x + k$

$\int e^{2x}\, dx$

$\dfrac{d}{dx}(e^{2x}) = 2e^{2x}$

$\dfrac{d}{dx}\left(\dfrac{e^{2x}}{2}\right) = \dfrac{\cancel{2}\, e^{2x}}{\cancel{2}} = e^{2x}$

so $\int e^{2x}\, dx = \dfrac{e^{2x}}{2}$

Inspection method

Find an anti derivative (or integral) of the following functions by the method of inspection.

1. $\sin 2x$
2. $\cos 3x$
3. e^{2x}
4. $(ax + b)^2$
5. $\sin 2x - 4e^{3x}$

(1) $\int \sin 2x \, dx$

$= -\frac{\cos 2x}{2}$

$\int \cos 3x \, dx$

$= \frac{\sin 3x}{3}$

$\int e^{2x} \, dx$

$= \frac{e^{2x}}{2}$

(4) $\int (ax+b)^2 \, dx$

$= \frac{(ax+b)^3}{3a}$

(5) $\int \sin 2x - 4e^{3x} \, dx$

$= -\frac{\cos 2x}{2} - 4\frac{e^{3x}}{3}$

Proper fraction

⇓

Degree of Numerator < Degree of Denominator

eg $\frac{x^1}{x^2+1}$ eg $\frac{x^2+1}{x^4+1}$

Improper fraction

Deg. of Numerator ≥ Degree of Denom.

eg $\frac{x^2}{x+1}$ eg $\frac{x^3}{x-1}$ eg $\frac{x^4}{x^2+1}$ eg $\frac{x^2-1}{x^2+1}$

Improper को Proper बनाओ

⇓

Long Division method

$D \overline{)\quad\quad}(Q$

$\overline{\quad R \quad}$

$Q + \frac{R}{D}$

13. $\displaystyle\int \frac{x^3 - x^2 + x - 1}{x - 1}\,dx$

Improper fraction

$$x-1 \,\big)\, \cancel{x^3} - \cancel{x^2} + x - 1 \,\big(\, x^2 + 1 \to Q$$

$x-1 \to D$

$$\begin{array}{r} \pm x^3 \mp x^2 \\ \hline x - 1 \\ \pm x \mp 1 \\ \hline 0 \to R \end{array}$$

$$= \int Q + \frac{R}{D}\,dx$$

$$= \int (x^2+1) + \frac{0}{x-1}\,dx$$

$$= \int x^2 + 1\,dx$$

$$= \frac{x^3}{3} + x + K$$

Q. $\int \frac{2x^3+4x^2+5x-1}{x+1}\,dx$

(Improper fraction)

$x+1 \,\overline{)\,2x^3+4x^2+5x-1\,}\,(2x^2+2x+3$

$\pm 2x^3 \pm 2x^2$

$2x^2+5x-1$

$\pm 2x^2 \pm 2x$

$3x-1$

$\pm 3x \pm 3$

$-4 \longrightarrow R$

$x+1 \to D$

$2x^2+2x+3 \to Q$

Now

$I = \int Q + \frac{R}{D}\,dx$

$= \int 2x^2 + 2x + 3 \;\; \frac{-4}{x+1}$

$= \frac{2x^3}{3} + \frac{2x^2}{2} + 3x - \frac{4\log|x+1|}{1}$

$+k$

thank

You